Abstract

Genome-wide association studies identify thousands of variant-trait associations, yet localizing true effector genes and resolving biological pleiotropy remains confounded by systemic cohort biases and linkage disequilibrium. Here, we present FALCON, a unified probabilistic model of direct genetic support, to construct the Genetic Map (G-Map)—a comprehensive, bias-adjusted catalog of human gene-trait associations. By correcting for redundant cohort signals, the G-Map reveals the true distribution of systemic pleiotropy across thousands of phenotypes. We systematically categorize this global genetic architecture into established clinical targets, rapid repurposing opportunities, and strictly novel discoveries. Our analysis highlights heavily burdened biological networks, demonstrating that developmental and immune-linked loci function as highly versatile pleiotropic hubs. By mapping the genetic basis of human disease from validated therapeutics to uncharted biological frontiers, the G-Map provides a robust, highly tractable foundation for novel target prioritization and drug repurposing.

Keywords: GWAS, Drug repurposing, Genetic architecture

1 Main

Human "genetic support" implicating genes in disease substantially increases drug target viability and provides a strong justification for experimental studies [Plenge, Minikel, Nelson]. Large-scale genome-wide association studies (GWAS) and exome-wide association studies (ExWAS) provide a strong foundation for this effort. These studies identify hundreds of thousands of variants associated with human traits. However, researchers struggle to determine the specific genes implicated by these associations. Common variant associations often fall in non-coding regions with unclear links to effector genes. Conversely, rare variant associations from ExWAS have limited statistical power. Furthermore, association studies typically report only highly significant associations. These stringent thresholds offer little information on the extent to which sub-significant associations provide genomic support for a gene and trait of interest.

Work to address these challenges mainly falls into three areas. First, effector gene prediction methods attempt to localize significant genome-wide association study (GWAS) signals to specific genes. Fine-mapping methods such as SuSiE[cite] localize GWAS signals to causal variants using sparse priors to calculate better posterior distributions. Fine-mapping is traditionally based on properties of variant association strength but can also be improved by leveraging functional annotations, including ATAC-seq[Corces, Bi], as demonstrated by tools like PolyFun[cite] and fgwas[cite]. These variants can then be linked to putative effector genes using variant-to-gene (V2G) scores that integrate multiple types of evidence (e.g. Hi-C[Jung], cS2G[Gazal]). These predictions are often manually combined into "effector gene lists" that accompany GWAS papers, although these lists lack standardization and rigor[Costanzo]. Second, gene-level approaches such as MAGMA[cite] and MetaXcan[cite] assign genetic support statistics to all genes in the genome by integrating multiple SNPs with (potentially) eQTLs. Third, data from ExWAS, in which burden[cite] or variance component tests[cite] aggregate heuristic masks[Trang] of variants can directly implicate disease-relevant genes, albeit with significantly less power than GWAS[19, Dornbos]. These data and methods each address various parts of the genetic support question, but their heterogeneous assumptions and mathematical frameworks make their integration difficult[HuGE].

To navigate this complexity, we developed a Bayesian statistical method named FALCON (Framework for Analysis of Linkage and Combined Omics using Networks). This model integrates association studies—taking into account correlation— with functional genomic datasets (functional annotations and V2G) to calculate genetic support. We built the network's architecture upon established genetic algorithms (e.g. SuSiE[cite]) and used empirically generated data as priors to inform this structure (e.g. LD Score regression[cite]). We validated the accuracy of this framework on both simulations and real data, using these results to identify the genetic and genomic features most useful for predicting genetic support. We applied FALCON to 1370 full GWAS summary statistics with a total of 117,748 curated GWAS lead SNPs to produce a Genetic Map (G-Map). We used this map to (a) conduct a retrospective analysis of drug target relative success; (b) uncover distinct tissue-mediated mechanisms between SNPs and genes; (c) and evaluate the utility of GWAS associations for prioritizing rare disease genes. This method and map provides a foundation to accurately estimate probabilities of genetic support between every gene and every complex trait.

2 Results

3 Overview of FALCON

FALCON is a probabilistic framework designed to model rare and common genetic associations and auxiliary epigenomic annotations as a function of an unknown variable quantifying a gene's genetic support. FALCON directly ingests standard association statistics while accounting for linkage disequilibrium and environmental noise, seamlessly integrating both common and rare variants by tailoring their underlying hyperparameter distributions. To inform its predictive priors, the model synthesizes diverse functional genomic annotations—including accessible chromatin, enhancers, and loss-of-function metrics—alongside sophisticated variant-to-gene (V2G) linkage scores derived from 3D DNA looping and eQTL mapping (see Supplemental Material). By anchoring on specific genes and evaluating them against all available genomic data and competing targets, FALCON introduces a "reverse genetics" approach that provides a foundation for identifying true causal associations.

In brief, FALCON models observed genetic association data, genomic annotations and linkage scores as a function of a primary latent variable for a gene's probability of disease relevance and secondary latent variables for variant probability, annotation relevance and linkage probability.

[Equation that shows the rigor used in FALCON here]

The primary output of this framework is gene genetic support—the posterior probability that a specific gene is functionally associated with a trait, conditioned on all observed data. This score—calculated via Gibbs sampling— serves as a standardized metric to evaluate genes within a high-dimensional phenotype space. Furthermore, FALCON's architecture simultaneously yields secondary outputs, including variant causal probability, fine-mapping, and fine-linking to identify the most likely target genes. Together, these probabilistic outputs generate an end-to-end map translating genetic variation into actionable insight to undercover functional biology (Fig. 1c).

We validated FALCON's architecture using simulations across diverse trait architectures and sample sizes. Based on Normalized Discounted Cumulative Gain (NDCG)[cite], FALCON's capacity to recover true disease-relevant genes was nearly optimal, reaching saturation at sample sizes where standard gene-level approaches continued to struggle. Implementation of the model in different scenarios demonstrated shows the importance of combined data. Removal of functional annotations caused a 60

Finally, we confirmed FALCON's performance on real-world data by evaluating ten UK Biobank traits—representing a broad heritability spectrum 10–55

4 The HUGEST- Map

To provide a catalog of human genetic support, we constructed the Genetic Map (G-Map) by applying FALCON on 1,370 bottom-line traits from the Knowledge Portal[cite] across 19,427 genes. The map establishes 208,854 Strong trait-gene associations, defined by a probability threshold of 0.34 based on the Jeffreys Scale with a 5.0

On average, individual traits are Strong associated with 152 genes. Conversely, 16,333 genes possess at least one Strong trait association, with 1,848 of these genes being associated with exactly one.

The network encompasses 118,177 distinct loci. The map identifies the nearest gene as the top candidate (the gene with the highest probability in the locus) on 73.38

Using relative success rate[cite], we calculated that genes with Strong genetic support from the G-Map are 3.50 times more likely to possess a functional drug target. Compared to the 2.70 rate from the baseline nearest-gene approach, this represents a 29.63

Using the curated list of effector genes in [Constanzo] and clinical information from open targets, The G-Map identified 203,214 novel gene-trait associations with and average of 148 novel genes per trait. From the subset of associations with no clinical trials, 16,380 have approved drug targets for the associated gene, making them prime candidates for rapid clinical repurposing. Dynamic Figure 3 and Figure 4.

By integrating the curated list of effector genes from [Constanzo] with clinical data from Open Targets, we systematically categorized the G-Map associations to funnel associations from established therapeutics down to novel discoveries. Within the clinical space, the network captures 1,555 established gene-trait associations. Expanding beyond active clinical trials into translational opportunities, we identified 16,380 associations involving genes with approved drug targets for other indications, rendering them prime candidates for rapid clinical repurposing. Finally, filtering the network for strictly uncharacterized targets uncovers an expansive frontier of 203,214 novel gene-trait associations, driven by an average of 148 novel genes per trait.

5 Coverage

To evaluate trait representation in the G-Map across established phenotypes, we performed a coverage analysis using the Mondo Disease Ontology [cite]. We employed PubMedBERT to calculate semantic similarity, mapping G-Map trait names directly to ontology terms. Applying a similarity threshold of 0.5 at level 3 of the ontology yielded a match success rate of 94.59

Analysis of the matched terms revealed distinct patterns in trait representation across the ontology. The categories exhibiting the highest proportional coverage were acute stress disorder and perinatal disease (both at 50

6 Overview of genetic architecture

To accurately capture the underlying genetic architecture and mitigate the redundancy and low power in GWAS cohorts, it is necessary to evaluate these associations through a bias-adjusted framework. Raw association counts often inflate genetic signals; thus, correcting for this bias is essential to represent a more accurate biological distribution of pleiotropy and polygenicity. By filtering out noise, this bias adjusted analysis shows the biologically active hubs across established clinical therapeutics, repurposable targets and novel discoveries.

The complete, bias corrected G-Map encompasses a network of 16,290 unique genes mapped across 1,152 distinct phenotypic traits, exhibiting a broad pleiotropy (N=7,393) where genes associate with an average of 1.5 traits. The map contains pleiotropic master regulators, with the most dominant genetic signals emerging from PBX2, NOTCH4, and TSBP1 (the three of them with 40 Strong associated traits). Phenotypically, global genetic burden is driven by undetermined stroke etiology (toastUNDETER; 1,872), complex lean mass phenotypes (TB-LM; 1,815), and platelet volume (PlatVol; 1,692). Consequently, the broadest categorical signals across the entire G-Map are deeply concentrated within hematological (17,223), musculoskeletal (4,575), and renal (4,177) domains.

Funneling this global architecture into the clinically validated space, the network captures 490 established therapeutic targets driving 299 distinct clinical phenotypes. The G-Map highlights a strong clinical inclination toward chronic inflammatory and structural conditions, where top genes: TUBB (4.42), TNF (2.78), and NR3C1 (2.69), and traits: ankylosing spondylitis (6.18), chronic obstructive pulmonary disease (COPD; 5.69), and TB-LM (4.84), modulate microtubule dynamics, systemic inflammation, and glucocorticoid signaling. Expanding beyond established indications, the repurposable class identifies [total—Genes_Repurposable] genes and [total—Traits_Repurposable] traits, providing a baseline for immediate translational opportunities. In this repurposable context, the strongest gene candidates are heavily anchored in immune regulation and the major histocompatibility complex, led by HLA-DRB1 (36.01), AGER (32.28), and HLA-DRB5 (30.43). The phenotypic signals and categories driving these repurposable opportunities closely mirror the global baseline.

Finally, filtering the map for novel associations uncovers [total—Genes_Novel] uncharted biological hubs across [total—Traits_Novel] distinct traits that have yet to be therapeutically exploited. Within this novel tier, the most dominant genes—PBX2 (40.57), NOTCH4 (40.56), and C6orf10 (40.09)—are identical to the global leaders, emphasizing that the bulk of the network's untapped pleiotropic potential resides in these developmental and immune-linked loci. The driving traits for these novel discoveries continue to be toastUNDETER (1,872.67), TB-LM (1,810.87), and PlatVol (1,692.64). This alignment firmly establishes the hematological (17,209.32), musculoskeletal (4,467.65), and renal (3,942.39) domains as the primary categories for new target discovery.